

1. General Description

The EM74LVC1G17 is a single buffer Schmitt-trigger. Inputs can be driven from either 3.3 V or 5 V devices. This feature allows the use of these devices as translators in mixed 3.3 V and 5 V environments. This device is fully specified for partial power down applications using I_{OFF} . The I_{OFF} circuitry disables the output, preventing the potentially damaging backflow current through the device when it is powered down.

2. Features and Benefits

- Wide supply voltage range from 1.65 V to 5.5 V
- Overvoltage tolerant inputs to 5.5 V
- High noise immunity
- ± 24 mA output drive ($V_{CC} = 3.0$ V)
- CMOS low power dissipation
- Latch-up performance exceeds 100 mA
- Direct interface with TTL levels
- I_{OFF} circuitry provides partial Power-down mode operation
- Unlimited rise and fall times
- Complies with JEDEC standard:
 - JESD8-7 (1.65 V to 1.95 V)
 - JESD8-5 (2.3 V to 2.7 V)
 - JESD8B/JESD36 (2.7 V to 3.6 V)
- ESD protection:
 - HBM ANSI/ESDA/JEDEC JS-001 Class 3B exceeds 8000 V
 - MM JESD22-A115C Class C exceeds 550 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options

EM74LVC1G17

Single Schmitt trigger buffer

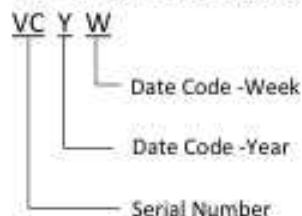
3. Ordering Information

Table 1. Ordering information

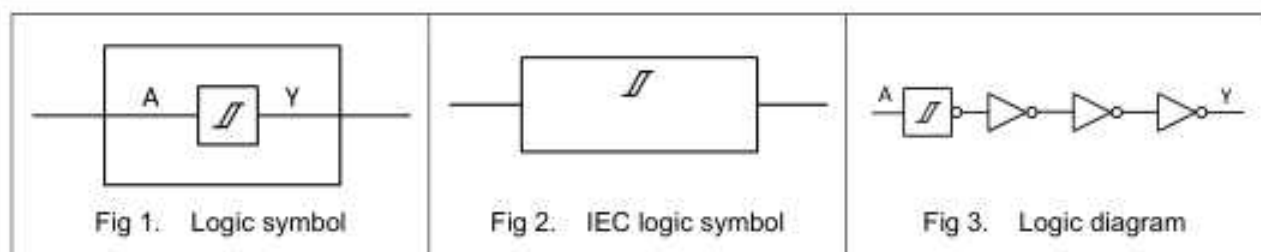
Type number	Topside marking	Package		
		Name	Description	Quantity
EM74LVC1G17GV	VCYW	SOT23-5L	SOT23 package, 5 pins 2.92 mm × 1.6 mm; 1.25 mm (Max) height	3000
EM74LVC1G17GW	VCYW	SOT353	SOT353 package, 5 pins 2.1 mm × 1.25 mm; 1.1 mm (Max) height	3000
EM74LVC1G17GS	VC	DFN1x1-6L	DFN1×1 package, 6 pins 1 mm × 1 mm; 0.42 mm (Max) height	3000
EM74LVC1G17GM	VCYW	DFN1x1.45-6L	DFN1.45×1 package, 6 pins 1.45 mm × 1 mm; 0.6 mm (Max) height	3000
EM74LVC1G17GX	VC	DFN0.8x0.8-4L	DFN0.8×0.8 package, 5 pins 0.8 mm × 0.8 mm; 0.4 mm (Max) height	3000

MARKING INFORMATION

NOTE: YW = Date Code.

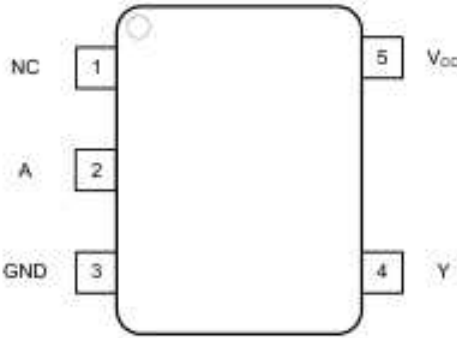
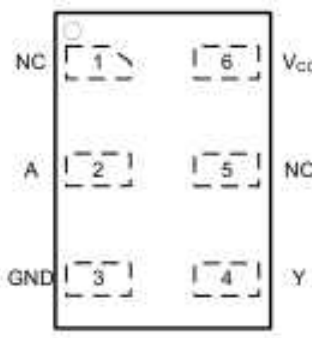
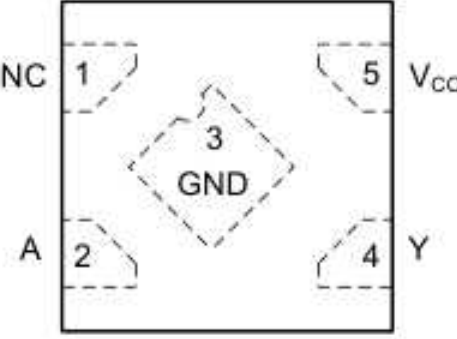


4. Function Diagram



5. Pinning Information

5.1. Pin map

 Fig 4. Top view pin configuration SOT23-5 and SOT353	 Fig 5. Top view pin configuration DFN6L
 Fig 6. Top view pin configuration DFN0.8x0.8-4L	

5.2. Pin description

Table 2. Pin description

Symbol	Pin			Description
	SOT23-5, SOT353 and DFN0.8x0.8-4L	DFN-6L	DFN-4L	
NC	1	1, 5	-	Not connected
A	2	2	1	Data input
GND	3	3	2	Ground (0V)
Y	4	4	3	Data output
V _{CC}	5	6	4	Supply voltage

6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

Input	Output
A	Y
L	L
H	H

7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	supply voltage		-0.5	6.5	V
I_{IK}	input clamping current	$V_I < 0$ V	-50		mA
V_I	input voltage	[1]	-0.5	6.5	V
I_{OK}	output clamping current	$V_O > V_{CC}$ or $V_O < 0$ V		± 50	mA
V_O	output voltage	Active mode [1]	-0.5	$V_{CC} + 0.5$	V
		Power-down mode; $V_{CC} = 0$ V [1]	-0.5	6.5	V
I_O	output current	$V_O = 0$ V to V_{CC}		± 50	mA
I_{CC}	supply current			100	mA
I_{GND}	ground current		-100		mA
P_{tot}	total power dissipation	$T_{amb} = -40$ °C to $+125$ °C		250	mW
T_{stg}	storage temperature		-65	150	°C

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{CC}	supply voltage		1.65		5.5	V
V_I	input voltage		0		5.5	V
V_O	output voltage	Active mode	0		V_{CC}	V
		Power-down mode; $V_{CC} = 0$ V	0		5.5	V
T_{amb}	ambient temperature		-40		125	°C

9. Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
V_{OH}	HIGH-level output voltage	$V_I = V_{TH}$ or V_{TL}						
		$I_O = -100\mu A$; $V_{CC} = 1.65 V$ to $5.5 V$	$V_{CC} - 0.1$			$V_{CC} - 0.1$		V
		$I_O = -4 mA$; $V_{CC} = 1.65 V$	1.2			0.95		V
		$I_O = -8 mA$; $V_{CC} = 2.3 V$	1.9			1.7		V
		$I_O = -12 mA$; $V_{CC} = 2.7 V$	2.2			1.9		V
		$I_O = -24 mA$; $V_{CC} = 3.0 V$	2.3			2.0		V
		$I_O = -32 mA$; $V_{CC} = 4.5 V$	3.8			3.4		V
V_{OL}	LOW-level output voltage	$V_I = V_{TH}$ or V_{TL}						
		$I_O = 100\mu A$; $V_{CC} = 1.65 V$ to $5.5 V$			0.10		0.10	V
		$I_O = 4 mA$; $V_{CC} = 1.65 V$			0.45		0.70	V
		$I_O = 8 mA$; $V_{CC} = 2.3 V$			0.30		0.45	V
		$I_O = 12 mA$; $V_{CC} = 2.7 V$			0.40		0.60	V
		$I_O = 24 mA$; $V_{CC} = 3.0 V$			0.55		0.80	V
		$I_O = 32 mA$; $V_{CC} = 4.5 V$			0.55		0.80	V
I_I	Input leakage current	$V_I = 5.5 V$ or GND ; $V_{CC} = 5.5 V$		± 0.1	± 1		± 1	μA
I_{OFF}	power-off leakage current	$V_{CC} = 0V$; V_I or $V_O = 5.5 V$		± 0.1	± 2		± 2	μA
I_{CC}	supply current	$V_I = 5.5V$ or GND ; $I_O = 0A$; $V_{CC} = 5.5V$		0.1	4		4	μA
ΔI_{CC}	additional supply current	per pin ; $V_{CC} = 2.3V$ to $5.5V$; $V_I = V_{CC} - 0.6V$; $I_O = 0A$		5	500		500	μA
C_i	input capacitance	$V_{CC} = 3.3V$; $V_I = GND$ to V_{CC}		5				pF

[1]All typical values are measured at $V_{CC} = 3.3V$ and $T_{AMB} = 25^\circ C$.

10. Dynamic Characteristics

Table 7. Dynamic characteristics

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 8.

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
t_{pd}	propagation delay	nA to nY; see Fig. 7 [2]						
		$V_{CC} = 1.65 \text{ V to } 1.95 \text{ V}$	4.0	10.6	19.5	4.0	19.8	ns
		$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$	2.5	5.9	10.6	2.5	10.9	ns
		$V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$	2.0	4.3	7	2.0	7.5	ns
		$V_{CC} = 4.5 \text{ V to } 5.5 \text{ V}$	1.5	2.9	4.8	1.5	5.0	ns
C_{PD}	power dissipation capacitance	per buffer ; $V_I = \text{GND to } V_{CC}$; $V_{CC} = 3.3\text{V}$ [3]		24				pF

[1] Typical values are measured at $T_{amb} = 25 \text{ °C}$ and $V_{CC} = 1.8 \text{ V, } 2.5 \text{ V, } 3.3 \text{ V and } 5.0 \text{ V}$ respectively.

[2] t_{pd} is the same as t_{PLH} and t_{PHL} .

[3] C_{PD} is used to determine the dynamic power dissipation (P_D in μW).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \sum (C_L \times V_{CC}^2 \times f_o)$ where:

f_i = input frequency in MHz;

f_o = output frequency in MHz;

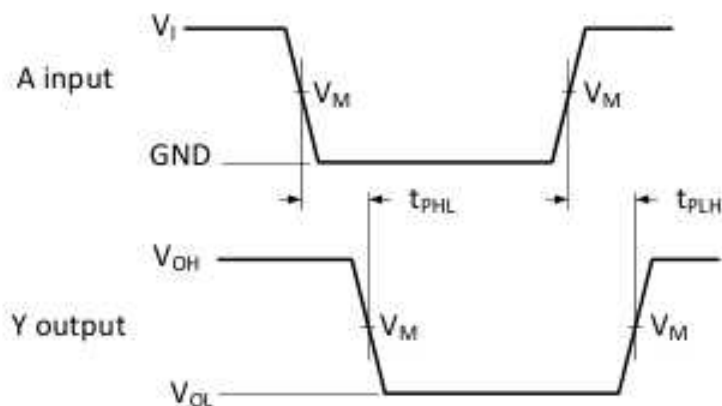
C_L = output load capacitance in pF;

V_{CC} = supply voltage in V;

N = number of inputs switching;

$\sum (C_L \times V_{CC}^2 \times f_o)$ = sum of outputs.

10.1. Waveforms and test circuit



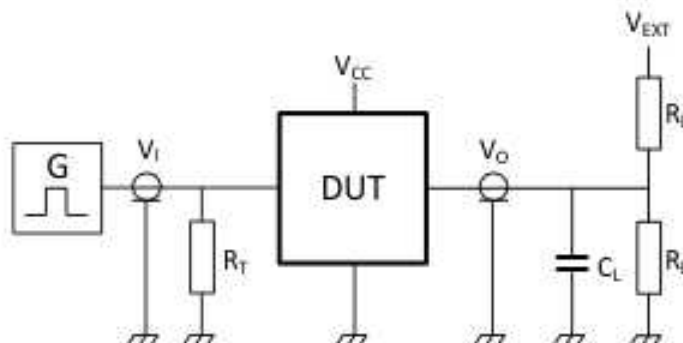
Measurement points are given in Table 8.

V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

Fig 7. The input nA to output nY propagation delays

Table 8. Measurement points

Supply voltage	Input	Output
V_{CC}	V_M	V_M
1.65 V to 1.95 V	$0.5V_{CC}$	$0.5V_{CC}$
2.3 V to 2.7 V	$0.5V_{CC}$	$0.5V_{CC}$
3.0 V to 3.6 V	1.5 V	1.5 V
4.5 V to 5.5 V	$0.5V_{CC}$	$0.5V_{CC}$



Test data is given in Table 9.

Definitions for test circuit:

R_L = Load resistance.

C_L = Load capacitance including jig and probe capacitance.

R_T = Termination resistance should be equal to the output impedance Z_o of the pulse generator.

V_{EXT} = External voltage for measuring switching times.

Fig 8. Test circuit for measuring switching times

Table 9. Test data

Supply voltage	Input		Load		V_{EXT}
V_{CC}	V_I	$t_r = t_f$	C_L	R_L	t_{PLH}, t_{PHL}
1.65 V to 1.95 V	V_{CC}	≤ 2.0 ns	30 pF	1 k Ω	open
2.3 V to 2.7 V	V_{CC}	≤ 2.0 ns	30 pF	500 Ω	open
3.0 V to 3.6 V	3 V	≤ 2.5 ns	50 pF	500 Ω	open
4.5 V to 5.5 V	V_{CC}	≤ 2.5 ns	50 pF	500 Ω	open

11. Transfer Characteristics

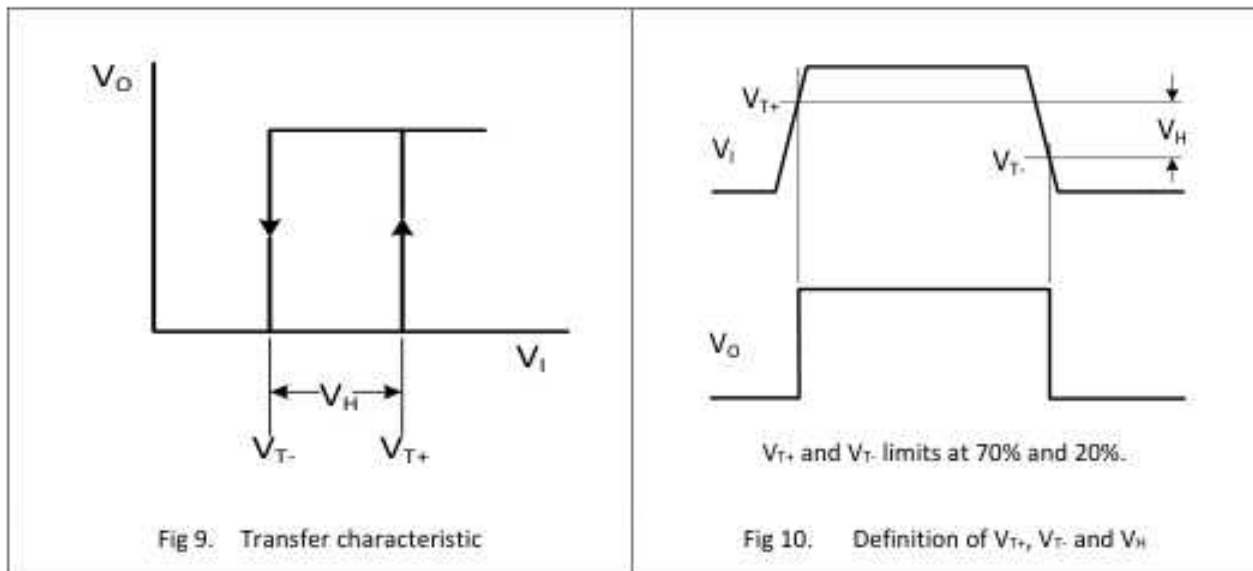
Table 10. Transfer characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
V_{T+}	positive-going threshold voltage	see Fig. 9 and Fig. 10						
		$V_{CC} = 1.8 \text{ V}$	0.82	1.0	1.14	0.79	1.14	V
		$V_{CC} = 2.3 \text{ V}$	1.03	1.2	1.40	1.00	1.40	V
		$V_{CC} = 3.0 \text{ V}$	1.29	1.5	1.71	1.26	1.71	V
		$V_{CC} = 4.5 \text{ V}$	1.84	2.2	2.36	1.81	2.36	V
		$V_{CC} = 5.5 \text{ V}$	2.19	2.6	2.79	2.16	2.79	V
V_{T-}	negative-going threshold voltage	see Fig. 9 and Fig. 10						
		$V_{CC} = 1.8 \text{ V}$	0.46	0.6	0.75	0.46	0.78	V
		$V_{CC} = 2.3 \text{ V}$	0.65	0.7	0.96	0.65	0.99	V
		$V_{CC} = 3.0 \text{ V}$	0.88	1.0	1.24	0.88	1.27	V
		$V_{CC} = 4.5 \text{ V}$	1.32	1.6	1.84	1.32	1.87	V
		$V_{CC} = 5.5 \text{ V}$	1.58	1.9	2.24	1.58	2.27	V
V_H	hysteresis voltage	see Fig. 9 and Fig. 10						
		$V_{CC} = 1.8 \text{ V}$	0.26	0.5	0.62	0.19	0.62	V
		$V_{CC} = 2.3 \text{ V}$	0.28	0.5	0.65	0.22	0.65	V
		$V_{CC} = 3.0 \text{ V}$	0.31	0.6	0.7	0.25	0.7	V
		$V_{CC} = 4.5 \text{ V}$	0.40	0.6	0.77	0.34	0.77	V
		$V_{CC} = 5.5 \text{ V}$	0.47	0.6	0.88	0.41	0.88	V

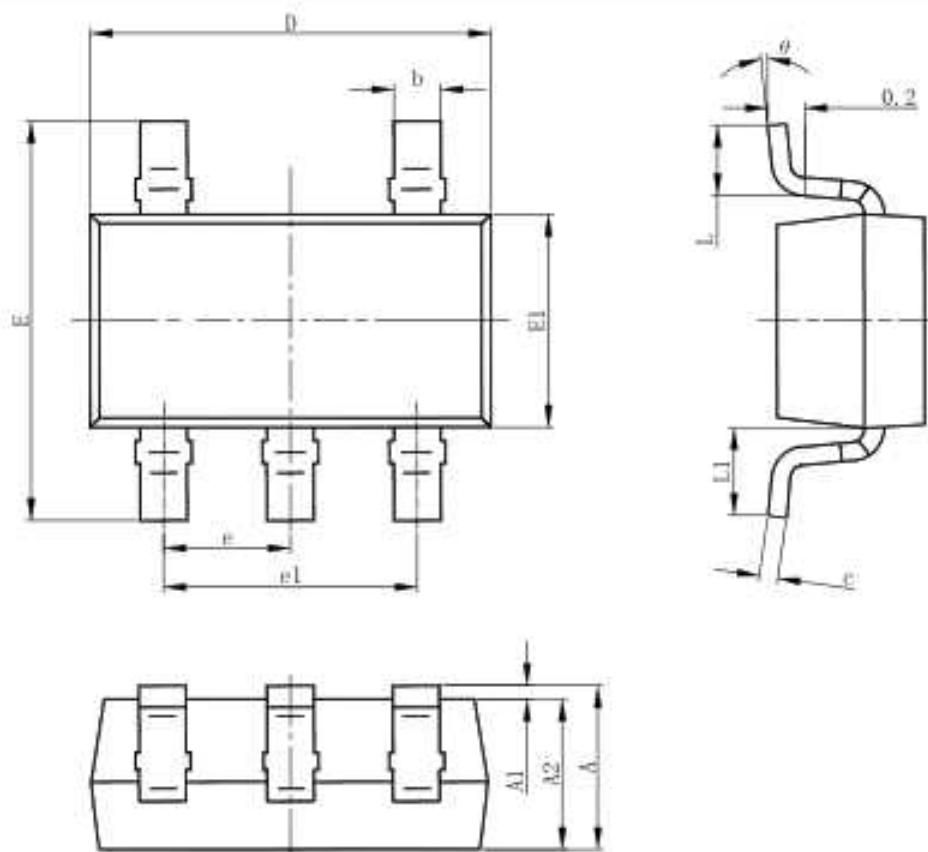
[1] Typical values are measured at $T_{amb} = 25 \text{ °C}$.

11.1. Waveforms transfer characteristics



12. Package Outline

SOT23-5L

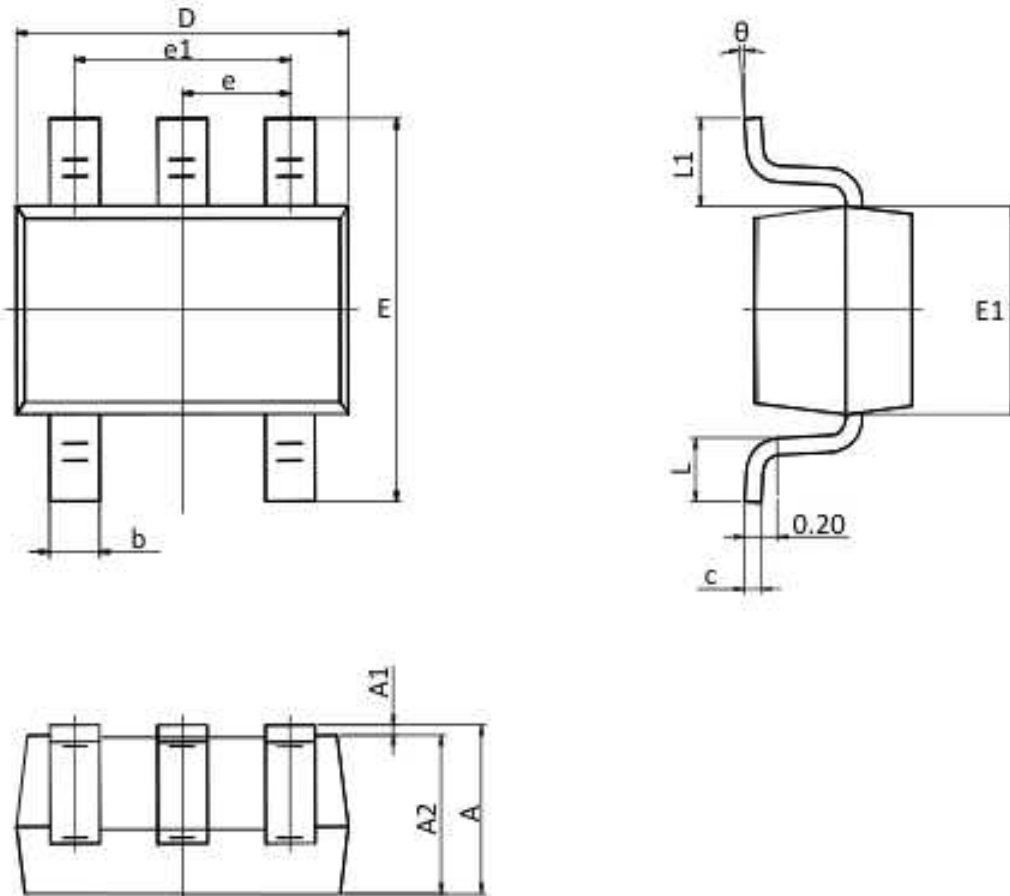


Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E1	1.500	1.700	0.059	0.067
E	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
L1	0.600REF.		0.024REF.	
θ	0°	8°	0°	8°

EM74LVC1G17

Single Schmitt trigger buffer

SOT353

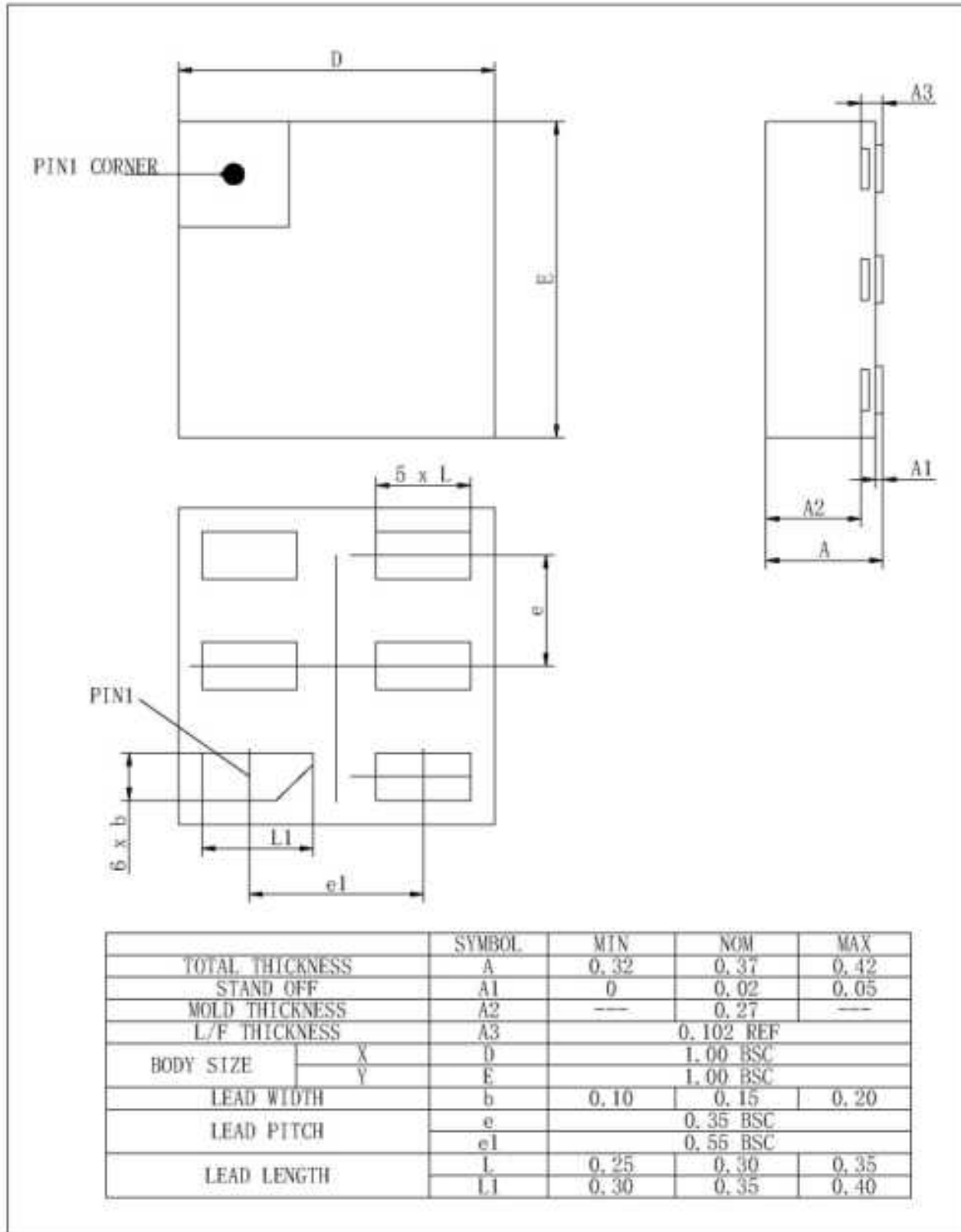


Symbol	Dimensions in Millimeters		Dimensions in Inches	
	Min.	Max.	Min.	Max.
A	0.900	1.100	0.035	0.043
A1	0.000	0.100	0.000	0.004
A2	0.900	1.000	0.035	0.039
b	0.150	0.350	0.006	0.014
c	0.110	0.175	0.004	0.007
D	2.000	2.200	0.079	0.087
E	2.150	2.450	0.085	0.096
E1	1.150	1.350	0.045	0.053
e	0.650 TYP.		0.026 TYP.	
e1	1.200	1.400	0.047	0.055
l	0.260	0.460	0.010	0.018
L1	0.525 REF.		0.021 REF.	
theta	0°	8°	0°	8°

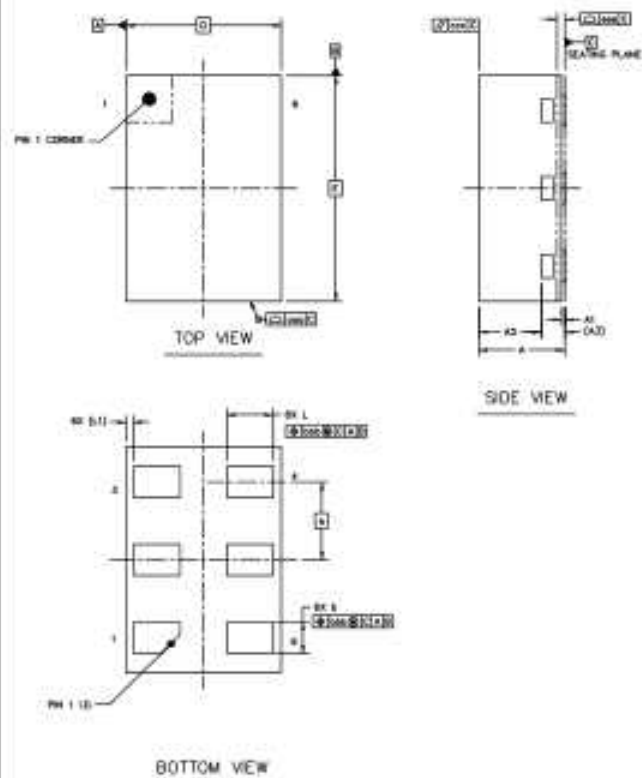
EM74LVC1G17

Single Schmitt trigger buffer

DFN1x1-6L



DFN1x1.45-6L



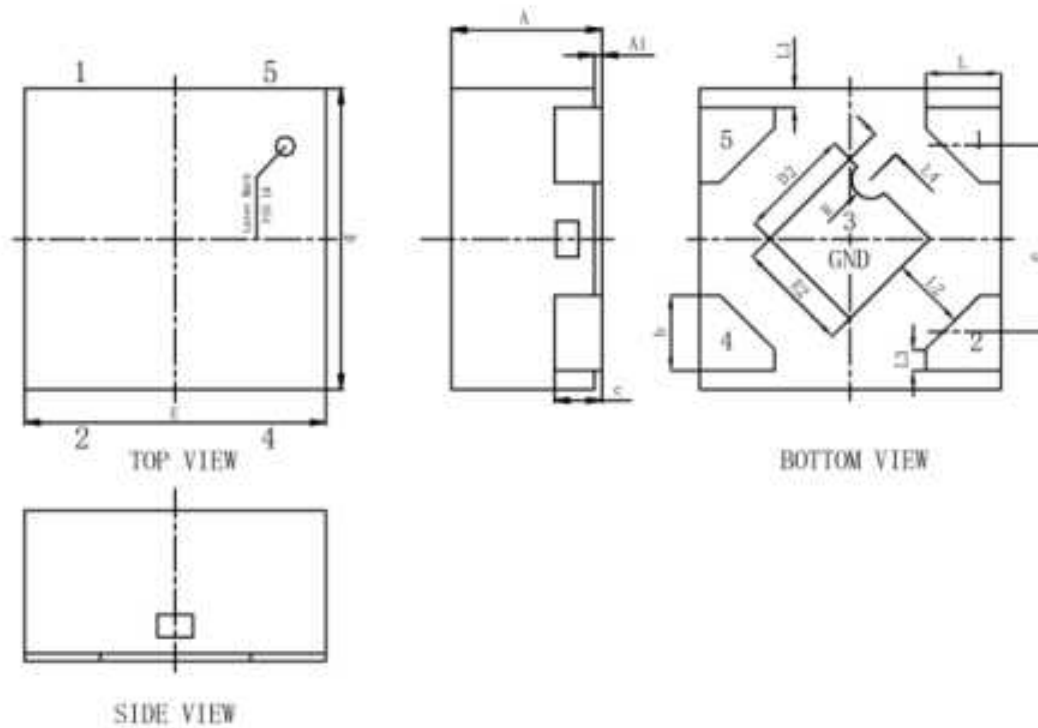
		SYMBOL	MFR	FORM	MAX
TOTAL THICKNESS		A	0.5	0.55	0.6
STAND OFF		A1	0	0.32	0.35
WELD THICKNESS		A2	0.005	0.4	0.005
L/T THICKNESS		A3		0.152 REF	
LEAD WIDTH		B	0.12	0.2	0.25
BODY SIZE	X	B		1 BSC	
	Y	B		1.45 BSC	
LEAD PITCH		C	0.5 BSC		
LEAD LENGTH	L	L	0.25	0.3	0.35
	L1	L1		0.08 REF	
PACKAGE EDGE TOLERANCE		000		0.1	
WELD FLATNESS		000		0.1	
COPLANARITY		000		0.05	
LEAD OFFSET		000		0.1	

NOTES:
1. REFER TO JDDC MG-130.
2. COPPLANARITY APPLIES TO LEADS, CORNER LEADS AND DIE ATTACH PWD.
3. BURN TO USE THE LEVEL 1 ENVIRONMENT-RELATED SUBSTANCE OF JEIT PRESCRIBING.
4. FINISH CLUTF = 50-204

EM74LVC1G17

Single Schmitt trigger buffer

DFN0.8x0.8-4L



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.35	—	0.40
A1	0.00	0.02	0.05
b	0.15	0.20	0.25
c	0.127REF		
D	0.75	0.80	0.85
D2	0.20	0.30	0.40
e	0.50BSC		
E	0.75	0.80	0.85
E2	0.20	0.30	0.40
L	0.15	0.20	0.25
L1	0.05REF		
L2	0.20REF		
L3	0.053REF		
L4	0.10REF		
R	0.05REF		

Package version	A0(mm)	B0(mm)	K0(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
SOT23-5L	3.23	3.17	1.37	0.25	4	8	4	Q3
SOT353	2.55	2.55	1.2	0.2	4	8	4	Q3
DFN1x1-6L	1.16	1.16	0.5	0.23	4	8	4	Q1
DFN1x1.45-6L	1.15	1.6	0.75	0.2	4	8	4	Q1
DFN0.8x0.8-4L	0.91	0.91	0.5	0.2	2	8	4	Q1

13.2. Reel and box dimensions

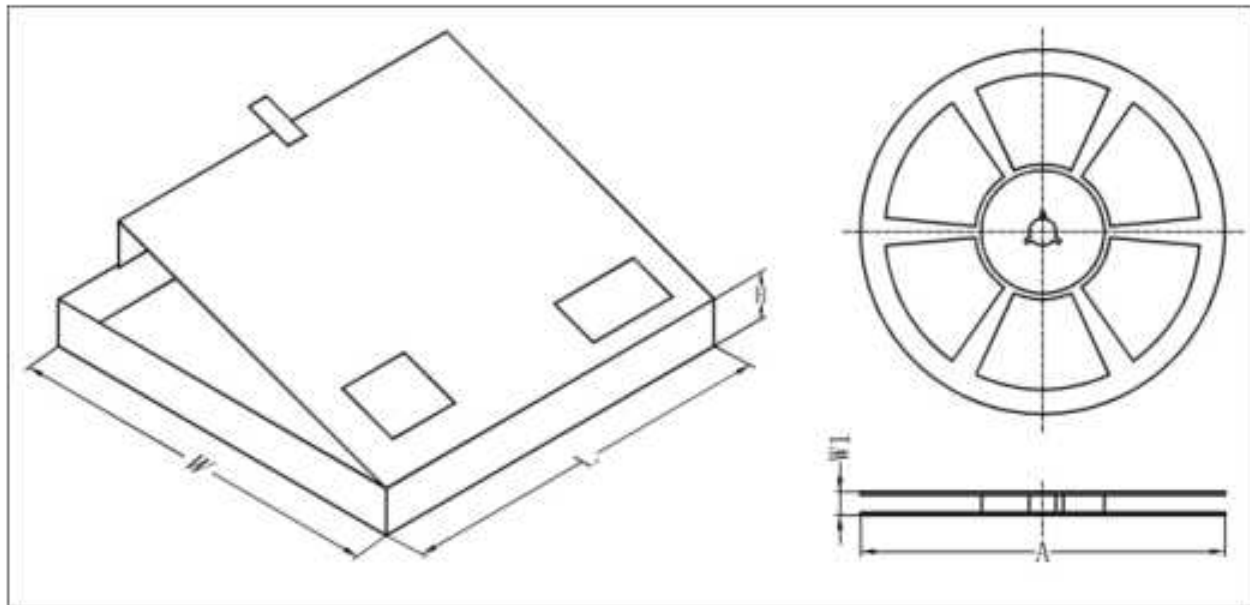


Table 12. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Reel Width W1 (mm)	SPQ (pcs)[1]	Reels per box	Outer box dimensions LxWxH(mm)[2]
SOT23-5L	GV	180	12.2	3000	1	210x200x40
SOT353	GW	180	12.2	3000	1	210x200x40
DFN1x1-6L	GS	180	12.2	3000	1	210x200x40
DFN1x1.45-6L	GM	180	12.2	3000	1	210x200x40
DFN0.8x0.8-4L	GX	180	12.2	3000	1	210x200x40

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

14. Abbreviations

Table 13. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	ElectroStatic Discharge
HBM	Human Body Model
MM	Machine Model
CDM	Charged Device Model
TTL	Transistor-Transistor Logic

15. Revision History

Table 14. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74LVC1G17 Rev1.4	Dec 20, 2025	Product datasheet		EM74LVC1G17 Rev1.3
Modifications:	Deleted type number EM74LVC1G17GX2 and EM74LVC1G17YZV.			
EM74LVC1G17 Rev1.3	Aug 30, 2025	Product datasheet		EM74LVC1G17 Rev1.2
Modifications:	Added type number EM74LVC1G17GX2 and package information.			
EM74LVC1G17 Rev1.2	Jul 14, 2025	Product datasheet		EM74LVC1G17 Rev1.1
Modifications:	Added tape and reel information.			
EM74LVC1G17 Rev1.1	Feb 20, 2025	Product datasheet		EM74LVC1G17 Rev1.0
Modifications:	Added DFN-4L package.			
EM74LVC1G17 Rev1.0	Oct 30, 2023	Product datasheet		